Volume I · Chapter 0 — Python & GitHub Fundamentals · Theory

> New **onboarding** chapter added after V1 — the entry point / foundations for the whole program (precedes Chapter 1). Hand-authored (not split from the curriculum) and maintained **independently** in the per-chapter tree. See `Volume-I-Chapter-0-context.md` for the AI interaction log.
>
> **Hour model (this chapter):** for every **1 h of theory** there are **2 h of labs** and **4 h of homework** (ratio **1 : 2 : 4**). Theory feeds later PowerPoint/slide + video generation.

Prerequisites: none — this is the on-ramp. Only a computer (Windows/macOS/Linux) and internet access are assumed.

Learning outcomes — the student can:

- Install and run Python; use the interpreter, run scripts, and work in Jupyter.
- Write correct Python: variables, types, operators, control flow, data structures, functions, modules/packages, files, and exception handling.
- Use `pip` and the standard library; read and navigate the official Python docs.
- Explain version control and use **Git** for the core workflow: `init`/`add`/`commit`/`status`/`log`, `.gitignore`, branches, merges, and conflict resolution.
- Use **GitHub**: repositories, clone, push/pull, the *Hello World* flow, and **pull requests**.

Topics (theory) — 10 h:

#	Topic	h
0.1	Setting up Python: install, the interpreter/REPL, editors (VS Code), Jupyter, running scripts	1
0.2	Syntax, variables, data types (numbers, strings, booleans)	1
0.3	Operators, input/output, comments, code style (PEP 8)	0.5
0.4	Control flow: `if`/`elif`/`else`, `for`/`while` loops, `break`/`continue`	1
0.5	Data structures: lists, tuples, dictionaries, sets	1.5

#	Topic	h
	(and comprehensions)	
0.6	Functions, arguments, scope; modules & packages; `pip`	1
0.7	Files (read/write) and exception handling; standard-library basics	1
0.8	Version control concepts; installing & configuring Git (`git config`)	0.5
0.9	Core Git workflow: `init`, `add`, `commit`, `status`, `log`, `.gitignore`	1
0.10	Branches, merging & resolving conflicts	0.5
0.11	GitHub: remotes, `clone`, `push`/`pull`, the *Hello World* flow, pull requests	1

Datasets/tools: Python 3 (system install or Anaconda), VS Code and/or Jupyter, `pip`, Git, a free GitHub account.

Assessment (theory): short formative quizzes — a Python-basics quiz (0.1–0.7) and a Git/GitHub quiz (0.8–0.11). Graded work lives in the Labs and Homework files.

Key decisions taught here:

- **Python install:** Anaconda vs. system Python vs. a `venv` — trade-offs for beginners.
- **Editor/runtime:** VS Code vs. Jupyter for different tasks.
- **Git habits:** commit granularity; when to branch; writing good commit messages.
- **GitHub auth:** SSH vs. HTTPS.

References & learning resources:

Official docs & tutorials

- W3Schools Python — https://www.w3schools.com/python/
- Python.org tutorial — https://docs.python.org/3/tutorial/index.html
- W3Schools Git — https://www.w3schools.com/git/
- GitHub "Hello World" — https://docs.github.com/en/get-started/using-github/hello-world

- GitHub — Git & GitHub learning resources — https://docs.github.com/en/get-started/start-your-journey/git-and-github-learning-resources
- Class Central — best Git/GitHub courses — https://www.classcentral.com/report/best-git-github-courses/

Curated GitHub repositories for Python practice (used mainly in Homework; verify current URLs)

- **30-Days-Of-Python** (Asabeneh) — step-by-step basics → web/data — `github.com/Asabeneh/30-Days-Of-Python`
- **Project-Based Learning** — build projects from scratch — `github.com/practical-tutorials/project-based-learning`
- **TheAlgorithms/Python** — 500+ algorithm/DSA implementations — `github.com/TheAlgorithms/Python`
- **Awesome Python** (vinta) — curated tools & frameworks — `github.com/vinta/awesome-python`
- **GeekComputers/Python** — real-world automation scripts — `github.com/geekcomputers/Python`
- **The Hitchhiker's Guide to Python** (Real Python) — best practices & setup — `github.com/realpython/python-guide`
- Also referenced (locate on GitHub, verify): *Learn-Python*, *Python Programming Exercises*, *Amazing Python Scripts*, *Learn Python 3*.
- Course tracks: *Python for Data Science, AI & Development* (IBM/Coursera), *Google IT Automation with Python*, *IBM AI Developer*.

  Note: several source links were provided as `lnkd.in` short-redirects; the canonical repo paths above are given where confident and should be re-verified before publishing.

Hours: Theory **10** + Labs **20** + Homework **40** = **70** (ratio 1 : 2 : 4).